Sensorless Torsion Control of Elastic-Joint Robots With Hysteresis and Friction

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***Abstract*—Sensorless torsion control is proposed for elastic-joint robots, with the geared drives subject to hysteresis and friction nonlinearities. The concept of the so-called "virtual torsion sensor" uses the motor torque and velocity, inherently available in most industrial robots, for estimating the reactive joint torque and predicting, based thereupon, the nonlinear joint torsion. The dynamics of the elastic-joint robot is described and augmented by a rate-independent torsion–torque hysteresis and nonlinear friction assumed on the side of motor drives. The classical two-degrees-of-freedom robot control, which includes the centralized torque feedforward control and proportional-derivative feedback control, is extended by the virtual torsion sensor in the loop. In particular, we show that the computed joint torsion can be injected in the feedback loop upon the motor position, thus making the control operating in a "virtual" joint output (link) space. The proposed control is experimentally evaluated on a single-joint setup consisting of an actuator with harmonic-drive gear and inertial load under additional impact of gravity.**

***Index Terms*—Elastic-joint robots, friction, hysteresis, motion control, nonlinearity, robot dynamics, robotics, torsion control.**

I. Introduction

THE challenges of controlling the elastic-joint robots are well known from the literature [1]–[5] and have continuously attracted the attention of researchers over the last three decades. Considerable contributions, above all from the viewpoint of analysis of the fast and slow dynamics of manipulators with elastic joints and their control, have been done in the second half of the eighties and in the nineties. Just to mention some of them, here are the works [6]–[11]. Also, an earlier contribution which provided experimental evidence of joint elasticities in the industrial robots and thus inspired redefining the motion control problems from the rigidly considered manipulators to that with joint elasticities can be found in [12]. Later, in two thousands, several theoretical and also experimentally evaluated works contributed further toward better understanding and regulating the flexible joint robots (see, e.g., [13]–[17]).

Various control theories have been applied for dynamics analysis and control design of elastic-joint robots. Among them are the singular perturbation theory (see, e.g., [6] and [9]), feedback (FB) linearization (see, e.g., [6] and [18]), energy shaping approach (see, e.g., [11] and [19]), and related passivity-based approaches (see, e.g., [17], [20], and [21]). Most of the previous methods, also those refereed previously, assumed the linear joint stiffness in the form of a torsional spring, which approximates the geared interconnection between the rotor inertia of the joint drives and the rigid-body inertia of the joint links. This simplifying assumption proved sufficient for a nearly linear range of the joint stiffness at relatively low operation torques and relatively low accuracy requirements posed on the link positioning. At the same time, the assumed linear stiffness made possible most of the performed analyses and control designs, since relying on powerful means of the classical control theory. However, when accepting the stiffening spring characteristics and hysteresis lost motion in elastic joints, which are the matter of fact in various geared robotic manipulators, the nonlinear torsion–torque relationship with certain state memory should be considered instead of a linear torsional spring. Although the presence of the aforementioned nonlinearities in elastic robotic joints, particularly those equipped with harmonic drives, have been known from former works (see, e.g., [22]–[24]), their practical consideration for modeling and control of elastic-joint robots found little attention in the past research. However, several explicit studies of harmonic-drive gear transmissions (see, e.g., [25]–[29]) revealed the experimental evidence of hysteresis in the torsion–torque characteristics, which can turn out as nonnegligible depending on the application. Further aspects of nonlinearities in elastic robotic joints can be found in, for example, [30] and [31]. Other works related to nonlinear reaction torques in variable-ratio gears in vehicles and kinematic errors of robot links due to the joint flexibility and measurement with motor encoders only can be found in [32] and [33], respectively.

This paper deals with elastic-joint robots whose modeling is extended through the torsion–torque hysteresis map and nonlinear friction in the joint drives. The based thereupon *sensorless torsion control* is motivated by the matter of fact that most of the industrial robot manipulators are operating by using the measured states which are available only on the motor side of the joint transmissions. At the same time, the rising requirements on positioning accuracy of the joint links and therefore payload, on the one hand, and the ongoing trends toward more lightweight and thus flexible structures and components, on the

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other hand, make the relative joint torsion a significant and challenging issue for control. The following contribution is based on the previous works [34]–[37], where the concept of observing the nonlinear joint torsion and using the related, so-called, *virtual sensor* (VS) in the loop has been introduced. The concept of VS of joint torsion has been equally discussed from the identification, application, and hysteresis-related viewpoints. The recent paper differs from the aforementioned previous works by the following: 1) describing uniformly the modeling and two-degrees-of-freedom control framework of multilink robotic manipulators; 2) addressing the related control design while incorporating the VS of joint torsion into the closed loop; and 3) showing a detailed experimental case study with positioning and torsion control which features an accuracy close to encoder's resolution. Furthermore, it is worth noting that the proposed method constitutes a concatenation of the estimation of reactive joint torque, based on the *generalized momenta* [38], [39], and the model-based prediction of joint torsion, based on the inverse hysteresis model. Therefore, the proposed method differs from the known "classical" nonlinear (disturbance) observers, e.g., [40] and [41].

The rest of this paper is organized as follows. Modeling of elastic-joint robots with geared drives subject to hysteresis and friction nonlinearities is described in Section II. The two-degrees-of-freedom control framework with the inclusion of the VS of the joint torsion is presented in Section III. In Section IV, the experimental case study is reported in detail. The concluding remarks and discussion on relevant issues for the future research are given in Section V.

II. Dynamic Model of Elastic-Joint Robots

The standard dynamic model of a serial-kinematic robotic manipulator assumes the following structure:

$$H(q)\ddot{q} + C(q,\dot{q}) + G(q) = u \tag{1}$$

where $q \in \mathbb{R}^n$ and $u \in \mathbb{R}^n$ are the vectors[1] of general joint coordinates, correspondingly driving torques which are directly supplied by the motor-based drives attached to the joints. The rigid-body dynamics is determined by the $H(q) \in \mathbb{R}^{n\times n}$ inertia matrix of the manipulator, which is augmented by the constant (state-independent) diagonal matrix elements of the motor drives' inertias. The vectors $C(q,\dot{q}) \in \mathbb{R}^n$ and $G(q) \in \mathbb{R}^n$ constitute the Coriolis/centrifugal and gravity torques correspondingly. Note that the parameterizing terms H, C, and G are all state dependent, thus rendering the robotic manipulator as nonlinear multivariable systems with cross-couplings between the single joints.

When accounting for joint elasticities, each of the overall n robotic joints assumes an additional degree-of-freedom so that the vector $\theta \in \mathbb{R}^n$ represents the angular joint input (motor) coordinates. Therefore, the extended robotic manipulator dynamics assumes $2n$ second-order differential equations as

$$H(q)\ddot{q} + C(q,\dot{q}) + G(q) = \tau \tag{2}$$

$$J\ddot{\theta} + \tau = u. \tag{3}$$

Here, $J = \mathrm{diag}(j_i) \in \mathbb{R}^{n\times n}$ is the positive diagonal matrix of the motor drives' inertias, and $\tau \in \mathbb{R}^n$ is the vector of joint torques. Note that (2) and (3) constitute a reduced dynamic model of elastic-joint robots, as it has been initially proposed in [6], without accounting for inertial couplings between the motors and links. This assumption is based on the fact that the angular kinetic energy related to the motor drives' rotors depends to the most part on their relative rotation only. This is justified by assuming the nominal gear ratios $N = \mathrm{diag}(n_i) \in \mathbb{R}^{n\times n}$ with $n_i \gg 1$ which relates the rigid-joint angular displacement to that of the motor drives by $\theta = Nq$. Recall that the angular velocity enters quadratically the corresponding kinetic energy, so that the own rotors' inertias differ from that due to inertial couplings by a quantity $\sim n_i^2$. Furthermore, we note that the dynamic model of elastic-joint robots, proposed in [6] and since then widely used in several works (see, e.g., [17] and the references therein), assumes the linear joint stiffness $K = \mathrm{diag}(k_i) \in \mathbb{R}^{n\times n}$ of torsional springs with $k_i > 0$, so that $\tau = K(N^{-1}\theta - q)$. In the following, we will denote the vector:

$$\Delta = N^{-1}\theta - q \tag{4}$$

by the *joint torsion* which reflects a relative displacement across the elastic joints, i.e., between the motor drives and joint link axes. Note that the joint damping, i.e., proportional to the $\dot{\Delta}$ quantity, and additional viscous (velocity-dependent) damping on the motor or link side can be equally included into the model. Also, to note is that the dynamic model (2) and (3) does not account for the external forces acting on the robotic manipulator which are, otherwise, entering the right-hand side of (2) through the corresponding manipulators' Jacobian.

In this paper, we extend the dynamic model (2) and (3) by two nonlinear terms which are the matter of fact for several geared robotic joints, particularly those which incorporate the harmonic drives. Recall that the harmonic drives are the compact, high-torque, and low-backlash gearing mechanisms often employed in the revolute robotic joints. However, due to a specific mechanical assembly, including a nonrigid flexspline, the harmonic drives are subject to substantial elasticities which are coupled with internal friction effects. For more details on nonlinearities in harmonic drives, we refer to the works [23], [26]–[28], [34], [35], [42], and [43].

The stiffening characteristics of elastic robotic joints based on harmonic drives can be captured by a polynomial function which, in a most simple case, includes the linear and cubic terms only. Thus, an elastic robotic joint exhibits a regressive nonlinear compliance as the relative torsion saturates at an increasing magnitude of the joint torque. Note that an appropriate elastic deformation range, i.e., without irreversible plastic deformation or even fracture, is considered. At the same time, the joint flexibility combined with an internal gearing friction and structural damping gives rise to a nonnegligible hysteresis in the torsion–torque characteristics. It is worth noting that, when

[1] In the following, and further on through the rest of this paper, we will denote all vector values by the small letters. Whenever required, a particular scalar value or a vector's element will be denoted by the subscript index, e.g., q_i or u_i. Furthermore, for the sake of simplicity, we will denote all dynamic quantities without time argument, except that this is required to emphasize an explicit dependence on the time argument, e.g., when comparing the dynamic and static behavior.

dealing with hysteresis phenomenon, a rate-independent torsion–torque map here is meant. That is the hysteresis trajectories depend explicitly on the amplitude and its history but not on the timescale on which these have been proceeded. For modeling the torsion–torque hysteresis of the ith joint, we apply the Bouc–Wen-type hysteresis model

$$\tau_i(\Delta_i) = \sum_j k_{i,j} \left(w_i \Delta_i^j + (1 - w_i) x_i^j \right) \tag{5}$$

(see [34] and [35]), where k_j, with $j = \{1, 3\}$, represents the linear and cubic stiffness coefficients. The weighting factor $0 < w_i < 1$ provides a relationship between the purely elastic ($w_i = 1$) and purely plastic ($w_i = 0$) hysteresis behaviors. The actual hysteresis behavior is captured by an internal state

$$\dot{x}_i = \phi_i \dot{\Delta}_i - \psi_i |\dot{\Delta}_i| |x_i|^{\eta_i - 1} x_i - \xi_i \dot{\Delta}_i |x_i|^{\eta_i} \tag{6}$$

where $\phi_i,\ \psi_i > 0, \xi_i \geq 0$, and $\eta_i \geq 1$ are the hysteresis control parameters. For more details on parameterization of the class of Bouc–Wen-type hysteresis models, we refer to [44].

To keep the modeling of motor drive friction simple as possible while capturing, at the same time, the most pronounced friction nonlinearities, we apply the steady-state Stribeck friction curve. Note that a more complex (dynamic) friction behavior includes the phenomenon of friction lag, also known as hysteresis in the velocity, and presliding hysteresis in displacement (see, e.g., [45] and [46] for details). Since our approach, however, aims at the torsion compensation at steady-state motion and final positioning, we neglect the more complex friction dynamics, this for the sake of an overall design and identification simplicity. The total friction torque of the ith motor drive is described by

$$f_i(\dot{\theta}_i) = \mathrm{sign}(\dot{\theta}_i) \Big[F_{c,i} + F_{s,i} \exp\Big(-V_i^{-\mu_i} |\dot{\theta}_i|^{\mu_i} \Big) \Big] + b_i \dot{\theta}_i. \tag{7}$$

Here, the standard Stribeck characteristic curve is parameterized by the Coulomb and Stribeck friction coefficients $F_{c,i} > 0$ and $F_{s,i} > 0$ correspondingly. The exponential parameters $\mu_i \neq 0$ and $V_i > 0$ are, respectively, the Stribeck velocity and shape factors; both determine the velocity weakening and strengthening curve. The linear viscous friction coefficient is denoted by b_i. It can be noted that, despite the static Stribeck friction model yields discontinuity at each motion reversal, i.e., velocity zero crossing, it does not affect the modeling purpose aimed by the virtual torsion sensor. On the contrary, the friction discontinuity at even transient impulses of the measured velocity provides a necessary persistent excitation of the joint torque estimator (see further in Section III-C), thus resulting in a faster convergence in vicinity to the steady state.

Capturing the hysteresis joint stiffness and motor drive friction as described previously, the overall dynamic model of an elastic-joint robot results in

$$H(q)\ddot{q} + C(q, \dot{q}) + G(q) = \tau(\Delta) \tag{8}$$

$$J\ddot{\theta} + \tau(\Delta) = u - f(\dot{\theta}). \tag{9}$$

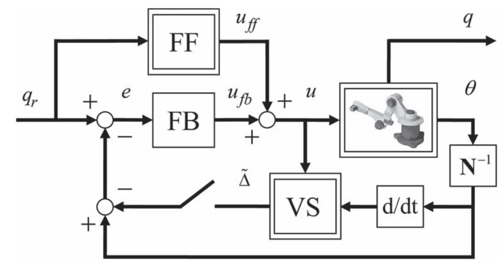

Fig. 1. Control framework of the elastic-joint robot with FF and FB controllers and virtual torsion sensor (VS).

Referring to both nonlinearities in (8) and (9), the following general remarks can be made before designing the control framework with a sensorless torsion compensation.

1) Since the joint torque hysteresis is a rate-independent $\Delta \mapsto \tau$ map, no dynamic impact on the eigen-behavior of the rigid manipulator, i.e., left-hand side of (8), occurs by extension. The torsion–torque hysteresis is an inherently dissipative element and provides an additional damping of manipulators' dynamics at the closed cycles. Hence, the $\tau(\Delta)$ hysteresis map can certainly reshape the frequency response of the elastic joint, comparing to the hysteresis-free stiffness case, but, at the same time, cannot significantly change its resonant characteristics. However, this assumption is made based on the general torsion–torque hysteresis properties and without a rigorous mathematical proof or experimental evidence. A more detailed analysis of elastic-joint dynamics with and without accounting for hysteresis is up to the future works.
2) The nonlinear friction appears in (9) as an input disturbance. It serves as an additional motor drive damping at steady-state motion and therefore can be respected when designing the FB control, as addressed further in Section III-B. The nonlinear Coulomb friction and related presliding friction at motion reversals are more challenging for linear FB control. In case of uncompensated Coulomb friction, the joint will stop within a dead zone around the desired position [14]. This matter of fact is well known from the analysis of the closed-loop controls in the presence of Coulomb friction (see, e.g., [47] and [48] for details). Nevertheless, one can show that increasing the proportional FB gain reduces the dead zone around the desired position. Since the focus of the recent work is on the sensorless torsion control of elastic-joint robots, we refrain from an explicit friction compensation in the following control design. However, in an experimental case study provided further in Section IV, we will show a possible improvement of the motor positioning accuracy when compensating for nonlinear Coulomb friction.

III. Two-Degrees-of-Freedom Control With Sensorless Torsion Compensation

A. Control Framework

One of the most widespread control architectures used for robotic manipulators is a two-degrees-of-freedom control as exemplary shown in Fig. 1. The feedforward (FF) control part assumes the dynamic manipulator model as in (1). Based thereupon, the inverse dynamics provides the desired drive torques u_{ff} which computation requires the reference robot trajectories to comply with $q_r(t) \in \mathcal{C}^2$. More details on the inverse dynamics control, respectively, FF torque control, can be found in the robotic literature, e.g., [2] and [5]. Here, we note that, when designing the FF control part, the joint elasticities are deliberately neglected, and only the rigid manipulator dynamics, i.e., the left-hand side of (2), is taken into consideration. Otherwise, as shown in [13], the computation of FF torque for an elastic-joint manipulator assumes the following form:

$$\hat{u}_{ff} = J\ddot{\theta}_r + H(q_r)\ddot{q}_r + C(q_r, \dot{q}_r) + G(q_r) \tag{10}$$

where

$$\ddot{\theta}_r = \ddot{q}_r + K^{-1}\Big[H(q_r)q_r^{(4)} + 2\dot{H}(q_r)q_r^{(3)} + \ddot{H}(q_r)\ddot{q}_r + \ddot{C}(q_r\dot{q}_r) + \ddot{G}(q_r)\Big]. \tag{11}$$

One can see that, for a complete FF torque computation, according to (10) and (11), the reference trajectory should comply with $q_r(t) \in \mathcal{C}^4$. Furthermore, we note that (11) assumes a linear joint stiffness, i.e., $\tau = K\Delta$. In order to stay by the general nonlinear joint torque $\tau(\Delta)$, we neglect the second right-hand side term (in brackets) of (11), which is related to the reference joint torsion. Therefore, we apply the reduced FF torque control as

$$u_{ff} = (J + H(q_r))\,\ddot{q}_r + C(q_r, \dot{q}_r) + G(q_r) \tag{12}$$

where the rigid manipulator dynamics is given by the nonlinear functions H, C, and G of the reference state and not of the actual state as in (1) and (2).

As has been formerly proposed and proven in [7] and after that shown and broadly discussed in the robotic literature (see, e.g., [2], [5], and the references therein), the simple proportional-derivative (PD) FB control can be used for both rigid and elastic-joint robots [13]. The PD control achieves asymptotic tracking of the desired joint position for which the steady-state error $e = q_r - q$ reflects in (compare with [2])

$$K_p e = G(q)$$

for the case of a rigid robot. Here, K_p is the diagonal matrix of positive proportional control gains. Note that, assuming the compensated gravity, the steady-state error $e = 0$ can be ensured. However, in case of an elastic-joint robot, the gravity will lead to a steady-state joint torsion with the correspondingly uncompensated residual positioning error. Note that, even if an explicit gravity compensation, like the FF one applied in this work, eliminates the steady-state position error in the motor space, i.e., $q_r - N^{-1}\theta = 0$, the undesired joint torsion leads inherently to $e \neq 0$.

The applied FB control law is

$$u_{fb} = K_p(q_r - N^{-1}\theta) + K_d(\dot{q}_r - N^{-1}\dot{\theta}) \tag{13}$$

where K_d is the diagonal matrix of positive derivative control gains. Here, the VS is first switched off, as in Fig. 1. Recall that the PD FB control operates in the motor space so that the motor states, reflected through the gear ratios, are fed back from the robotic manipulator. The overall control vector, according to the block diagram in Fig. 1, is

$$u = u_{ff} + u_{fb} \tag{14}$$

while both control parts are operating independently from each other. The nonlinear FF control mainly aims at decoupling the complex manipulator dynamics based on the nominal inverse dynamic model. Furthermore, it allows for faster transients and hence reference tracking in virtue of accounting for inertias of the manipulator and motor drive rotors. The linear FB control aims at compensating for the model uncertainties, low disturbances, and inaccuracies of capturing the initial states of the multivariable robotic system.

The VS of joint torsion receives the vectors of the drive torques and velocities as input and provides the vector of predicted torsion $\tilde{\Delta}$ as output. The latter injects the measured motor positions in the loop so that, if $\tilde{\Delta} = \Delta$, then the PD FB control becomes operating in the "virtual" joint output (joint link) space. In Section III-C and D, we will describe the VS in more detail and show the stability of its inclusion into the FB control loop. The latter is required since the $\tilde{\Delta}$ values are computed in an FB manner from the measured motor torque and velocity states.

In the following, we will denote the motor states reflected through the gear ratios solely by θ and $\dot{\theta}$ while keeping in mind that the nominal gear reduction N^{-1} is already included into the computation, correspondingly measurement.

B. Design of PD Feedback Control

When designing the PD feedback control, we assume that the state-dependent manipulator dynamics and related cross-couplings between the single joints are mostly decoupled by the FF control (12). Therefore, the PD FB control becomes decentralized since each ith joint can be considered separately as a SISO system. Under this assumption and when neglecting first the impact of hysteresis and friction nonlinearities, the single elastic joint can be approximated by

$$\hat{H}_i\ddot{q}_i - k_i\Delta_i - d_i\dot{\Delta}_i = 0 \tag{15}$$

$$J_i\ddot{\theta}_i + b_i\dot{\theta}_i + k_i\Delta_i + d_i\dot{\Delta}_i = u_i \tag{16}$$

where $\hat{H}_i$ is the linearized inertia of the ith manipulator link. The parameters k_i and d_i are the linear joint stiffness and damping coefficients, and b_i is the viscous damping of the motor drive. The latter approximates the velocity-dependent friction, apart from the motion reversals. Note that, for the rest of this section, we will skip the subindex i, this for the sake of simplicity.

Here, it is worth to remark that the local linear model (15) and (16) discloses high fidelity with the measured frequency response of real elastic joints, when providing sufficiently large excitation amplitudes and, at the same time, sufficiently small relative displacements around the point of linearization. This matter of fact will be demonstrated further in Section IV-B. Similar experimental observations can be found, for example, in [12].

Rewriting the linear model (15) and (16) into the Laplace domain with argument s results in

$$P_l(s)q(s) = O(s)\theta(s) \tag{17}$$

$$P_m(s)\theta(s) = u(s) + O(s)q(s) \tag{18}$$

where

$$P_l(s) = \hat{H}s^2 + ds + k \tag{19}$$

$$P_m(s) = Js^2 + (b+d)s + k \tag{20}$$

are the forward transfer functions of the joint link and motor drive correspondingly. The forward and feedback couplings between both are given by

$$O(s) = ds + k. \tag{21}$$

One can easily show that the open-loop transfer function is

$$\frac{\theta(s)}{u(s)} = P_l(s)\left(P_l(s)P_m(s) - O^2(s)\right)^{-1}. \tag{22}$$

Introducing the transfer function of PD FB control as

$$C(s) = K_d s + K_p \tag{23}$$

one obtains the closed-loop transfer function

$$\frac{\theta(s)}{q_r(s)} = C(s)P_l(s)\left(P_l(s)P_m(s) + C(s)P_l(s) - O^2(s)\right)^{-1}. \tag{24}$$

When writing out the characteristic polynomial of the transfer function (24), one will see that this is of the fourth order and has bulk coefficients composed of the complete model and control parameters. Finding the roots of such polynomial and hence eigenvalues of the closed loop in an analytic form can be hardly solvable. However, one can show that, for physically reasonable model parameters, i.e., positive masses, damping, and stiffness, the characteristic polynomial remains Hurwitz and thus asymptotically stable independent of the choice of positive control gains. In fact, merely two control parameters do not allow for a fully solvable pole assignment. Therefore, the first conjugate-complex pole pair, related to the joint elasticities, remains with an imaginary (oscillatory) part independent of the PD control gain selection. The PD control gains allow assigning efficiently only another (second) pole pair, also to be real poles. At the same time, the first conjugate-complex pole pair migrates within the complex s-plane and that toward the open-loop zeros (cf., with [2, Ch. 6.5]).

Referring to the practical pole assignment, the close-loop bandwidth should be selected quite far below the natural frequency of the elastic joint. Note that the joint's natural frequency is determined by

$$\omega_0 = \sqrt{\frac{k}{M}} \quad \text{with} \quad M = (J + \hat{H})^{-1} J \hat{H} \tag{25}$$

where M is called the fast inertia (see [1] for details). It is self-evident that keeping the control bandwidth far from the joint's natural frequency allows avoiding as possible the excitation of the joint's resonance mode. Recall that the resonant joint behavior can provoke the undesirable link vibrations and disturb the positioning control at the operation.

C. VS of Joint Torsion

The VS of joint torsion is briefly summarized in this section, while for more details, we refer to [35] and [36]. The VS of joint torsion assumes two main parts which can be designed, correspondingly identified, separately. At the same time, both are serially connected so that the output of the first serves as the input of the second. Thus, the error propagation at predicting the joint torsion $\tilde{\Delta}$ is a composite mapping since $(u, \dot{\theta}) \mapsto \tilde{\tau} \mapsto \tilde{\Delta}$. However, since the errors are neither integrative nor multiplicative, the uncertainties of both parts yield acceptable for predicting the joint torsion.

The first part of VS aims at providing the estimate of the joint torque $\tilde{\tau}$ based on the dynamic modeling of the motor drive and the method refereed as generalized momenta (see [38] and [39] for details). Introducing the vector of generalized momenta $p = J\dot{\theta}$, one can rewrite the motor drive dynamics (9) as

$$\dot{p} = u - f(\dot{\theta}) - \tau. \tag{26}$$

Once the drive and friction torque vectors are assumed to be known/computable from the system measurements, one can capture the estimate of generalized momenta dynamics by

$$\dot{\tilde{p}} = u - f(\dot{\theta}) - r \tag{27}$$

where the residual vector $r = L(\tilde{p} - p)$ is assumed to be proportional to the error vector of generalized momenta. Here, $L > 0$ is the diagonal matrix of error gains which is the design parameter. Note that (27) is a standard observer for a class of nonlinear systems, with the nonlinear term as a function of measurable outputs [39], here with nonlinear friction f. Furthermore, it can be shown that since

$$r = L\left(\int \left[u - f(\dot{\theta}) - r\right] dt - p\right) \tag{28}$$

the residual state dynamics complies with

$$\dot{r} + Lr = L\tau. \tag{29}$$

From (29), one can see that the residual state r follows the unknown joint torque while featuring a linear exponentially stable dynamics. Therefore, the estimated joint torque can be assumed as $\tilde{\tau} = r$ while being lagged behind the real torque value τ by the vector of time constants L^{-1}.

The second part of the VS assumes the nonlinear model of torque transmission as in (5). Since the modeled joint torque contains a static, i.e., purely elastic, contribution (denoted further by α) and dynamic, i.e., plastic, contribution (denoted further by β), the overall torsion–torque hysteresis map, according to (5), can be rewritten as

$$\tau(\Delta, t) = W\alpha(\Delta) + (I - W)\beta(\Delta, t) \tag{30}$$

where $W = \mathrm{diag}(w_i) \in \mathbb{R}^{n \times n}$ is the diagonal matrix of weighting factors and I is the identity matrix. Since the static inverse α^{-1}-term is available and the dynamic β-term can be provided in the integral form, that according to (6), one can compute the joint torsion as

$$\tilde{\Delta}(\tilde{\tau}, t) = \alpha^{-1}\left(W^{-1}\left[\tilde{\tau} - (I - W)\beta(\tilde{\Delta}, t)\right]\right). \tag{31}$$

D. Impact of VS Inclusion

In order to analyze the impact of including the VS into the closed control loop, consider the $\tilde{\Delta}$ value as though acting directly upon the input torque vector u. Therefore, the predicted torsion occurs as an input disturbance vector for which the transfer function can be written as

$$\frac{u\left(\tilde{\Delta}(s)\right)}{\tau(s)} = K^{-1}K_p(L^{-1}s + I)^{-1} \tag{32}$$

according to (29) and proportional FB gain K_p. Note that, since the observer bandwidth should be not too high, which requires low L gain values, the FB coaction of $\tilde{\Delta}$ is restricted to the proportional gain only, i.e., without the derivative part. Furthermore, we note that, for the transfer function analysis, the linear joint stiffness here is assumed. From (32), one can recognize that an arbitrary joint torque dynamics is low-pass filtered as long as $L^{-1} > 0$. Recall that L is equally a control design parameter. It is also worth saying that the low-pass observer characteristics, concatenated with the proportional FB of torsion state, will reshape the closed-loop pole configuration. Here, however, we solely notice that an appropriate combination of L and K_p provides a stable VS inclusion realizable based on analysis of the pole configuration.

The closed-loop steady-state behavior under the impact of $\tilde{\Delta}$ can be examined by returning to (18) and substituting there the following control:

$$u(s) = C(s)(q_r(s) - \theta(s)) + K^{-1}K_p(L^{-1}s + I)^{-1}\tau(s). \tag{33}$$

Taking the steady-state limit $s \to 0$ results in

$$K\theta = K_p(q_r - \theta) + K_p(\theta - q) + Kq. \tag{34}$$

Eliminating in (34) the equal terms, one obtains

$$K(\theta - q) = K_p(q_r - q). \tag{35}$$

It is evident that, since $\theta = q$ at steady state, here without gravity, (35) will hold for link positioning error $q_r - q = 0$.

IV. Experimental Case Study

A. Single-Link Elastic Robotic Joint

The following experimental case study is accomplished on the laboratory setup which is adequate for emulating a single-link elastic robotic joint with nonlinearities under consideration. The experimental setup is schematically shown in Fig. 2. The joint actuator, based on the torque-controlled permanent magnet synchronous motor, integrates the harmonic-drive gear with elasticities and motor encoder as well. The inertial load disc is augmented by an additional weight block which produces the position-dependent gravity load. For identification

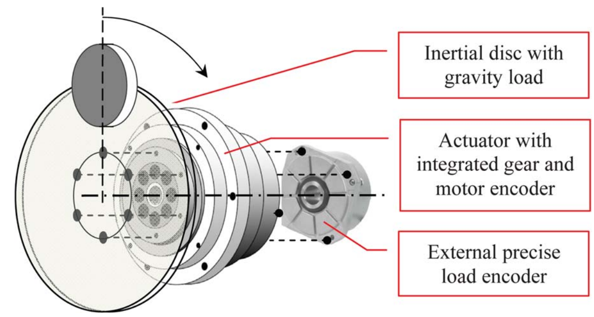

Fig. 2. Schematic representation of the used experimental setup with elastic joint and additional gravity load.

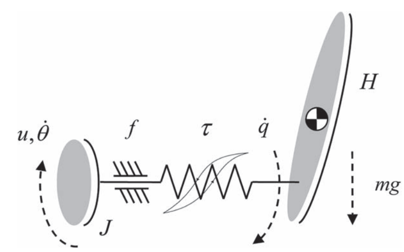

Fig. 3. Mechanical structure of the single-link elastic robotic joint.

and evaluation purposes, an external high-resolution output encoder is applied through the hollow shaft on the load side of the actuator. Recall that the load (joint link) sensing is not used for control. All signals are real time, while the sampling rate of the dSpace-based control unit is set to 2 kHz. For further details on the experimental setup, we refer to [35].

The principal mechanical structure of a single-link elastic robotic joint is shown in Fig. 3. Note that this constitutes a typical configuration of the second or third axes of a standard serial-kinematic robotic manipulator with rotary joints. There, the configuration-dependent joint loads are under strong impact of gravity torques produced by subsequent elements of the kinematic chain and additional payload applied to the end-effector as well. For the sake of simplicity and due to the limits of available hardware, we consider the load inertia H as constant and thus configuration independent. Also, no Coriolis and centrifugal torques, as in (8), appear in this case. Therefore, the overall motion dynamics from (8) and (9) can be reduced to

$$H\ddot{q} + G(q) = \tau(\Delta) \tag{36}$$

$$J\ddot{\theta} + \tau(\Delta) = u - f(\dot{\theta}) \tag{37}$$

where the gravity term $G = mgl\sin(q)$ is parameterized by the lumped mass m, gravity constant g, and lever arm l to the center of gravity (see Fig. 3). Note that, for the rest of Section IV, we will use the same terms of dynamics equations, while meaning the scalar values instead of the vectors and matrices considered before.

B. System Identification

The system identification based on the measured experimental data is accomplished in three following steps.

First, the linear local model (17) and (18) is determined in frequency domain by using the measured frequency responses

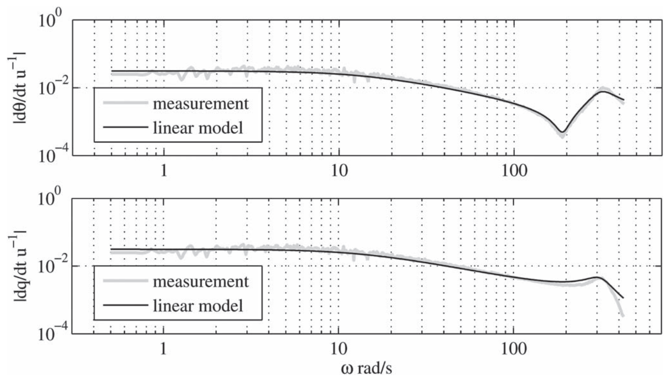

Fig. 4. Measured frequency responses and identified linear local model; frequency response of motor velocity (above) and link velocity (below).

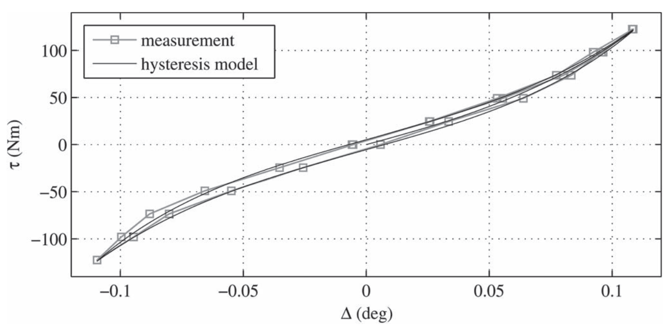

Fig. 5. Statically measured torsion–torque characteristic points and identified (based thereupon) hysteresis model of the elastic joint.

from the input torque to the motor velocity and to the link velocity (see Fig. 4 above and below correspondingly). The measurement is realized by applying an additionally (low-pass) filtered random binary signal excitation in the range 0.1–100 Hz. Note that the relative link position remains close to 0° so that a low impact of gravity torque can be practically neglected. The frequency response of the identified linear model, equally shown in Fig. 4, coincides well with the measurements.

Next, the hysteresis of elastic-joint transmission (5) is fitted in the least-squares sense on the statically measured torsion–torque characteristic points, both shown in Fig. 5. Recall that, since the hysteresis map is rate-independent, the interpolated data between the single measured points can be proceeded at an arbitrary timescale. For more details on the static torsion–torque measurements and related identification procedure of determining the hysteresis parameters, we refer to [35] and [36].

Having the identified motor drive inertia and torsion–torque hysteresis map, and, respectively, its inverse, the nonlinear friction $f(\dot{\theta})$ is identified from the controlled motion experiment, the same as will be further used in Sections IV-C–E for the control evaluation. The measured motor drive velocity and joint torsion, both used as the inputs of the inverse dynamic model (37), are depicted in Fig. 6(a) and (b) correspondingly. The measured input torque, depicted in Fig. 6(c), is used for the least-squares fitting of the nonlinear friction parameters. The corresponding inverse dynamic model response is shown in Fig. 6(c) over the measurement. The determined steady-state Stribeck curve of nonlinear friction (7) is shown in Fig. 7 as a function of the relative motor drive velocity.

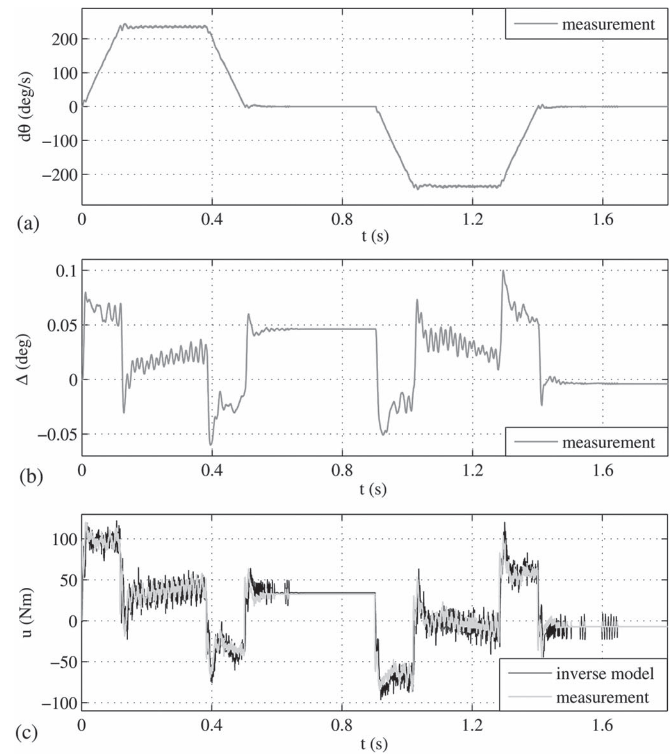

Fig. 6. Motion profile used for identification of motor friction. (a) Measured motor velocity. (b) Measured joint torsion. (c) Measured input torque and over that computed by the inverse dynamic model.

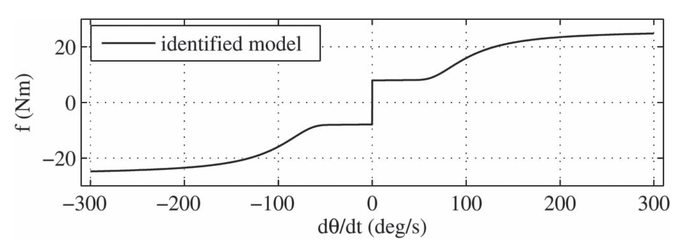

Fig. 7. Identified friction model of the steady-state Stribeck curve.

Furthermore, we note that the gravity term $G(q)$ from (36), which is required for the design of the FF controller but not of the VS, has been identified from an additional set of controlled low-velocity steady-state experiments. For more details on gravity identification, we refer to [36].

C. Trajectory Tracking Control

Using the identified local linear model (17) and (18), the PD FB control (23) is designed by evaluating the poles of the closed-loop system (24). The determined pole configuration is shown in Fig. 8 opposite to the poles of the identified plant model. Note that the dominant pole has the angular frequency of 113 rad/s, while the conjugate-complex poles, related to the joint elasticities, have the angular frequency of 222 rad/s. The latter is below the natural frequency of 313 rad/s of the elastic joint. The determined high-gain control is quite stiff and ensures, in combination with the FF control part, a fast position transient toward the steady state. The applied control parameters are listed in Table I. Note that the identified model and related calculus, correspondingly control design, are with the

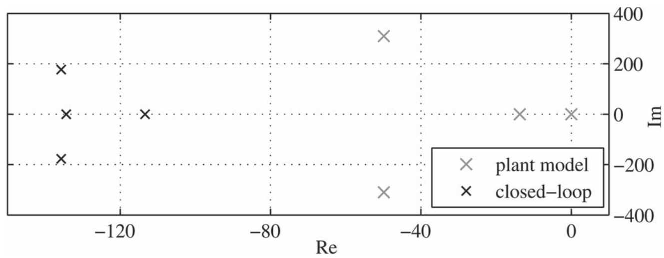

Fig. 8. Pole diagram of the identified local linear plant model (17) and (18) and closed-loop system with determined PD FB control.

TABLE I
CONTROL PARAMETERS USED FOR EXPERIMENTAL EVALUATION

K_p	K_d	L
18000	345	50

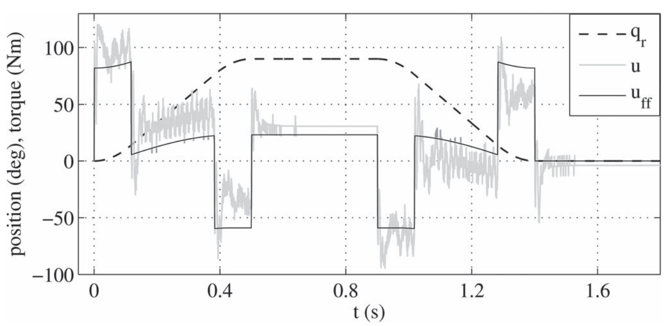

Fig. 9. Evaluated motion trajectory; reference link position versus overall control value (measured) and FF control part (computed).

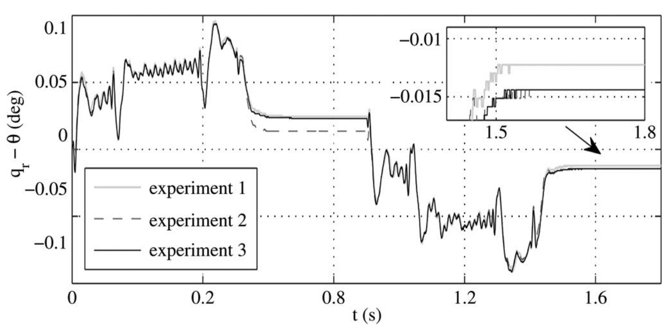

Fig. 10. Control error of motor position tracking (repeated experiments).

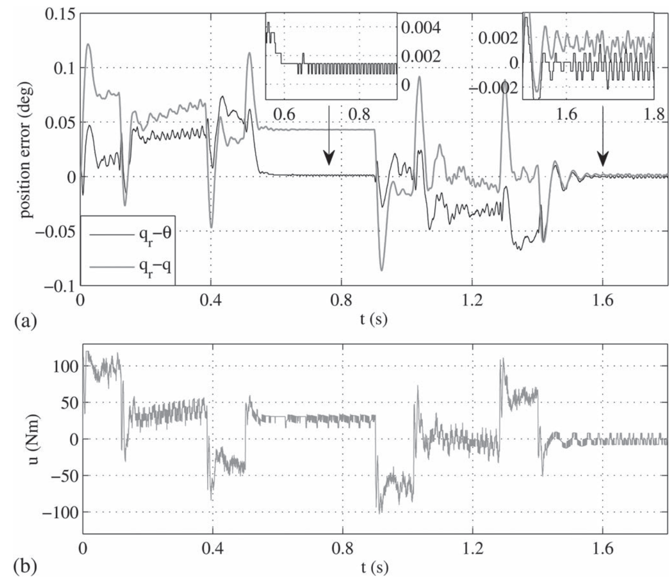

Fig. 11. (a) Control error of motor and link position tracking and (b) corresponding control value when using the robust control extension (38). Zoom-in inclusions show the steady state.

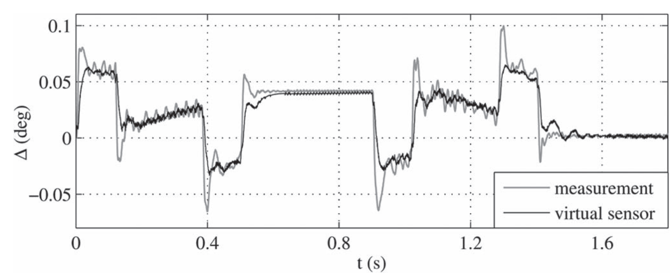

Fig. 12. Measured and predicted joint torsion when using control (38).

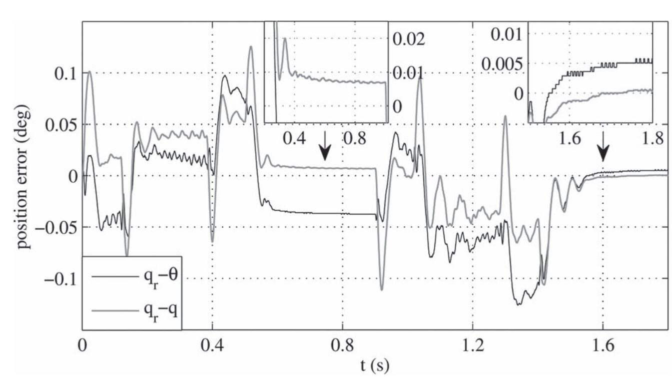

Fig. 13. Control error of motor and link position tracking when using the VS-based torsion compensation and robust control extension (38). Zoom-in inclusions show the steady state.

"radians" and "radians per second" units, while the results shown in Figs. 5–7 and 9–14 are with the "degrees" and "degrees per second" units, transformed for the sake of comprehensibility.

The motion trajectory under evaluation, shown in Fig. 9, is designed by using the standard linear segment with parabolic blends (LSPB) algorithm (see, e.g., [2] for details). Recall that the LSPB algorithm provides $\mathcal{C}^2$ reference trajectories, thus requiring the finite motor drive accelerations. The designed motion trajectory constitutes a fast displacement of the load from 0° to 90°, which is the position with maximal gravity, and after an idle state a return to 0°, and that at the same reference velocity. From the joint torsion point of view, the upper position provides the maximal steady-state torque conditions, while the lower position, correspondingly, the minimal. The measured control value is equally shown in Fig. 9, together with the computed FF control part.

The positioning errors of the motor drive obtained from three repeated experiments are shown in Fig. 10. While the overall trajectory tracking error is quite low, one can see that the residual steady-state motor positioning errors are present for both—upper (90°) and lower (0°) positions of the load. The residual steady-state errors vary between the repeated experiments and are not fully reproducible as can be seen from Fig. 10 (see the zoom-in inclusion). This is related to nonlinear presliding, equally as stiction, friction which remains uncompensated by

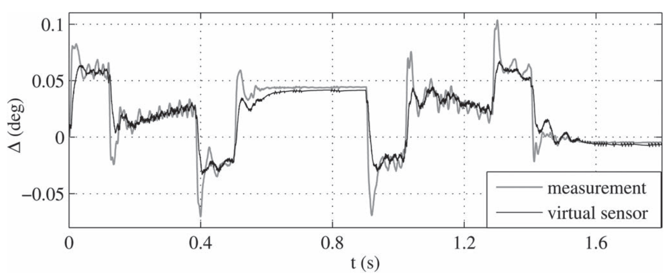

Fig. 14. Measured and predicted joint torsion when using the VS-based torsion compensation and robust control extension (38).

the two-degrees-of-freedom motion control (14). Here, it is worth to accentuate that including an additional integral control part cannot be always successful for improvement of the steady-state positioning accuracy within presliding friction range. For explicit studies and experimental evaluations of that, we refer to [49] and [50]. Since the magnitude of the residual steady-state positioning errors lies in a range close to that of the joint torsion, it appears necessary to eliminate the former as possible, so as to be able to adequately evaluate the torsion compensating method. In the following, we apply a simple robust control extension which allows minimizing the steady-state motor positioning error.

D. Robust Control Extension

Since an explicit compensation of steady-state motor positioning errors is out of scope in this work, we solely apply a simple robust control extension

$$u_r = K_r \text{sign}(q_r - \theta + \lambda\dot{\theta}) \tag{38}$$

which is of sliding-mode type. An analysis and evaluation of the sliding-mode control for compensating the presliding friction during positioning can be found, for example, in [51]. Here, we solely note that the gain $K_r = 7$ has been assumed to be of the same level as the static Coulomb friction coefficient. Furthermore, the sliding surface parameter $\lambda = 80$ has been determined from analyzing the $(\theta, \dot{\theta})$ phase portrait of the closed-loop control system evaluated previously, as reported in Section IV-C. Using the robust control extension (38), the overall control law (14) results in

$$u = u_{ff} + u_{fb} + u_r. \tag{39}$$

The same reference position as in Fig. 9 has been evaluated when applying the control (39). The control error of both, motor drive and link position tracking, are shown opposite to each other in Fig. 11(a). The corresponding control value is depicted in Fig. 11(b). One can see that the steady-state motor positioning error is significantly reduced through the robust control extension, comparing to Fig. 10. The residual errors of about 0.001° are is the range of the encoder's resolution, as can be seen from the zoom-in inclusions in Fig. 11(a). At the same time, one can see that, for lower load position (0°), the robust control extension induces certain oscillatory transient and residual chattering. This is inherent for involving a discontinuous control part (38) but negligible here since the residual chattering is in the range of the encoder's resolution. The difference between the motor and link positioning errors corresponds to the actual joint torsion. The latter is shown in Fig. 12 together with the predicted joint torsion, i.e., the output of VS. One can see that the predicted joint torsion is in accord with measurements, not only for the upper and lower load positioning but equally for the steady-state velocities and accelerations, after the related transients.

E. VS-Based Torsion Compensation

When allowing for the torsion compensation by using the VS, the fed back position and, respectively, FB control are adjusted by switching on the $\tilde{\Delta}$ signal, as from Fig. 1. The same motion trajectory as before has been evaluated. The tracking control errors of both, motor drive and link positions, are shown in Fig. 13, here also with zoom-in inclusions for the steady states. Obviously, the motor position error is now shifted, comparing to that in Fig. 11, since the FB control (with robust extension) is now operating on the $\theta - \tilde{\Delta}$ quantity. At the same time, the link position error is clearly reduced comparing to that in Fig. 11. The zoom-in inclusions in Fig. 13 show that the link position error at the upper load position (90°) lies below 0.01° and that at the lower load position (0°) is close to zero. The transients of link position error remain nearly the same as in the uncompensated case, compared with Fig. 11. At the same time, one should note that the accuracy of torsion compensation relies strictly on the exactness of VS prediction. The latter is shown in Fig. 14 in comparison to the measured joint torsion. One can see that both curves coincide well for the lower load position, while they slightly differ at steady state of the upper load position. It should be noted that the shown steady-state error depends on the initial values of both encoders, which are not absolute encoders, and thus on the capturing of the initial value of the torsional hysteresis. We comment on this essential issue in the following concluding remarks.

V. Concluding Remarks and Discussion

In this paper, we have described and evaluated with experiments the sensorless torsion control of elastic-joint robots. The standard dynamic model of elastic-joint robots has been extended by the rate-independent torsion–torque hysteresis map and nonlinear friction. The friction has been assumed on the side of the motor drives which actuate the joints with elasticities. The two-degrees-of-freedom control structure, which includes the centralized FF control based on the inverse dynamic model of the rigid-link manipulator, and PD FB control of the single joints has been summarized and addressed from the design point of view. The so-called VS of joint torsion, which constitutes the core of the proposed method, has been described while incorporating the joint torque estimation, which is based on the generalized momenta, and inverse torque–torsion hysteresis map. The inclusion of VS into the closed control loop has been analyzed and discussed regarding the stability and steady-state positioning error of the joint links. We have shown a detailed experimental case study accomplished on a single joint setup under gravity. The studied joint is actuated by a servo motor with harmonic-drive gears and exhibits well-detectable hysteresis and friction nonlinearities. The experimental system identification has been shown as required for the control design. A significant improvement in the link positioning accuracy could be achieved with the designed and applied VS of joint torsion in the control loop.

One of the most relevant issues related to the practical evaluation and operation of VS is the initial state of torsion–torque hysteresis. Since no absolute encoders have been used for evaluation, the relative torsion captured at the initial state is inherently biased despite zero control and zero gravity torques. That impact we have observed by repeating the same control experiments and evaluating the torsion prediction versus the measurements. For operating the VS with hysteresis, it can be important first to drive the joint in a particular torsion–torque state where the hysteresis memory is erased and thus the modeled hysteresis can be properly initialized. Thinkable is applying a short impulse of maximal input torque, which would be propagated to the joint torque but without inducing an apparent motion of the motor drive and joint link. At the same time, the rate-independent hysteresis would be saturated, which allows a well-defined hysteresis state (see, e.g., [37] for details). A more detailed analysis and evaluation of the initial value problem for elastic robotic joints with hysteresis is up to the future works.

There are also other interesting research issues related to the model (8) and (9). Among are the impact of nonlinear torque with hysteresis on the resonant characteristics of the manipulator and motor drive dynamics, particularly on the joint damping. Furthermore, an accurate motor positioning control can require an explicit compensation of nonlinear friction, as, for example, in [50] and [52], since this became visible from the experiments shown in Section IV-C and D. Moreover, the robustness of VS and overall (extended) control framework versus the parameter uncertainties appears as significant for the future research. The presented approach is also intended to be evaluated under time-varying operation conditions and with respect to the parameter uncertainties. Here, the manipulator-related restrictions posed on the model identification should be also accounted.

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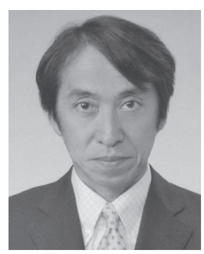

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